

HAT-P-12b

R-1213

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_220207 · created 2026-07-31T07:59:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459617.9239 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.012
Transit depth (Rp/Rs)^2	2.23 +/- 0.35 [%]
Orbital Inclination (inc)	87.68 +/- 0.84
Airmass coefficient 1 (a1)	1.208 +/- 0.015
Airmass coefficient 2 (a2)	-0.148 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	None
Optimal Aperture	3.89
Optimal Annulus	11.64
Transit Duration (day)	0.0894 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.012 vs 0.1406 (deviation 0.69σ)
Duration fit vs published	0.0894 d vs 0.0975 d (deviation 1.09σ)
Mid-time O–C	0.1 min
Depth SNR	12.3
Residual scatter	1.25 %

Light curve

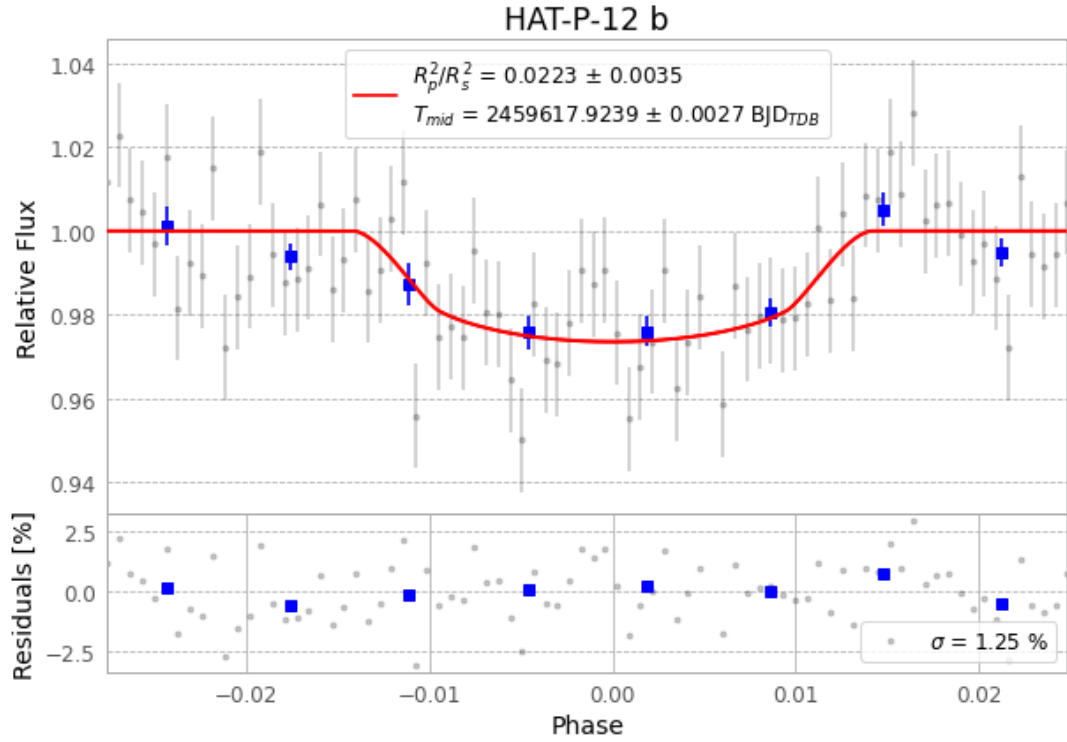


Figure 1 — FinalLightCurve_HAT-P-12 b_2022-02-07.png (SHA-256 2b00bdeb193232d8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [227, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8b9f5069e3314c83...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), HATP-12220207075714.FITS → HATP-12220207120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-220207.log (86a94cdebb00...), FinalLightCurve_HAT-P-12 b_2022-02-07.png (2b00bdeb1932...), FinalLightCurve_HAT-P-12 b_2022-02-07.csv (12911575277b...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 07:59Z.